

CANONICAL ROOT RINGS IN HIGHER DIMENSIONS: RIGIDITY, GCD FORESTS, AND ISOLATED ASYMPTOTICS

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ABSTRACT. For the Geigle–Lenzing graded ring $T(\mathbf{p}) = \mathbb{C}[X_1, \dots, X_n]/(\sum_i X_i^{p_i})$, let $\vec{\omega}$ be its dualizing element and let $\delta\vec{\tau} = \vec{\omega}$. We study its root section ring $R(\mathbf{p}, \vec{\tau}) = \bigoplus_{m \geq 0} T_{m\vec{\tau}}$ in dimension at least three. A hypersurface of weights $(a_1, \dots, a_n; h)$ satisfies

$$h - \sum_i a_i = \delta, \quad \frac{h}{\prod_i a_i} = \left(\frac{1 - \sum_i 1/p_i}{\delta} \right)^{n-2}.$$

For $n \geq 4$, a hypersurface root ring is rigid: $N(1 - \sum_i 1/p_i) = \delta$, so only the maximal root occurs. Among all rings satisfying this rigid maximal-root equation, we classify the complete intersections: they are exactly those for which the pairwise gcds form a multiplicative ultrametric. The canonical laminar forest gives the Hilbert basis, regular sequence, embedding dimension, primitive weights, and the hypersurface subclass as its zero- or one-node cases. Finally, on the full isolated locus we prove

$$\sum_{\delta \leq X} I_n^{\text{iso}}(\delta) \sim C_n X (\log X)^{n-1}$$

with an explicit Euler-product constant and its exact parity split. This yields multiplicity $\gg_n (\log \delta)^{n-1}$ along an unbounded sequence of indices.

1. INTRODUCTION

Throughout the paper $n \geq 3$. Let $\mathbf{p} = (p_1, \dots, p_n)$ be a sequence of integers $p_i \geq 2$ such that

$$(1.1) \quad \nu(\mathbf{p}) := 1 - \sum_{i=1}^n \frac{1}{p_i} > 0.$$

The Brieskorn–Pham ring

$$(1.2) \quad T = T(\mathbf{p}) = \mathbb{C}[X_1, \dots, X_n]/\left(\sum_{i=1}^n X_i^{p_i}\right)$$

is graded by the rank-one abelian group

$$(1.3) \quad L = L(\mathbf{p}) = \frac{\bigoplus_{i=1}^n \mathbb{Z}\vec{x}_i \oplus \mathbb{Z}\vec{c}}{(p_i\vec{x}_i - \vec{c})_{1 \leq i \leq n}}, \quad \deg X_i = \vec{x}_i.$$

For $n = 3$ this is the coordinate grading of a weighted projective line in the sense of Geigle and Lenzing [4]. For arbitrary n , the pair (T, L) is a Geigle–Lenzing hypersurface in the higher-dimensional sense of Herschend–Iyama–Minamoto–Oppermann [6, Section 2.1]. In this hypersurface case its dualizing element is [6, Proposition 2.10]

$$(1.4) \quad \vec{\omega} = \vec{c} - \sum_{i=1}^n \vec{x}_i.$$

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The canonical section ring $\bigoplus_{m \geq 0} T_{m\vec{\omega}}$ is a higher-dimensional analogue of a triangle singularity; canonical Veronese subrings in this setting also appear in [6, Section 3.4]. Its hypersurface case was studied in [5], extending the surface constructions of Dolgachev and Wagreich [2, 10]. More precisely, in the canonical case $\delta = 1$ and under the hypersurface hypothesis for $n \geq 4$, [5, Theorem 1.2] proves finiteness, [5, Proposition 2.9] gives $N\nu = 1$, [5, Proposition 3.2] proves that isolatedness is equivalent to $R = T$, [5, Theorem 1.3] gives the isolated Diophantine equation, and [5, Theorem 1.4] gives $a(R) = 1$. The root-specific results below extend these statements from the canonical ring to integral roots $\delta\vec{\tau} = \vec{\omega}$.

The root criterion and finite-invariant description in Sections 2–3 are also needed in our simultaneously prepared three-variable companion paper [3]. They are included in both papers, with the same proofs, so that each is logically self-contained. The companion develops the automatic Cox normal form and divisor classification for $n = 3$. Beyond these shared preliminaries and the canonical case of [5], the new results here are the extension of hypersurface rigidity to arbitrary integral roots, the gcd-forest complete-intersection classification, and the isolated defect-height asymptotics for $n \geq 4$.

We replace $\vec{\omega}$ by a positive integral root

$$(1.5) \quad \delta\vec{\tau} = \vec{\omega}, \quad \delta \in \mathbb{Z}_{>0},$$

and consider the root section ring

$$(1.6) \quad R(\mathbf{p}, \vec{\tau}) = \bigoplus_{m \geq 0} T_{m\vec{\tau}}.$$

Put

$$(1.7) \quad N = \text{lcm}(p_1, \dots, p_n), \quad k = N\nu, \quad d_{\max} = \max\{e > 0 : e \mid k, \gcd(e, N) = 1\}.$$

The possible roots are completely arithmetic.

Theorem 1.1 (Canonical roots). *There is a unique $\vec{\omega} \in L$ such that $\vec{\omega} = d_{\max}\vec{\omega}$. For a positive integer δ , the following are equivalent:*

- (1) *there is a $\vec{\tau} \in L$ with $\delta\vec{\tau} = \vec{\omega}$;*
- (2) *$\delta \mid k$ and $\gcd(\delta, N) = 1$;*
- (3) *$\delta \mid d_{\max}$.*

When they hold, the root is unique and $\vec{\tau} = (d_{\max}/\delta)\vec{\omega}$.

The Cartier dual of $L/\mathbb{Z}\vec{\tau}$ is a finite diagonal group $G_{\vec{\tau}}$, and $R = T^{G_{\vec{\tau}}}$. Molien's formula then determines the canonical shift and the leading Laurent coefficient of the Hilbert series. In particular, if

$$R \cong \mathbb{C}[z_1, \dots, z_n]/(f), \quad \deg z_i = a_i, \quad \deg f = h,$$

then

$$(1.8) \quad h - \sum_i a_i = \delta, \quad \frac{h}{\prod_i a_i} = \left(\frac{\nu}{\delta}\right)^{n-2}.$$

Extending the canonical case of [5, Proposition 2.9], our first root-specific higher-dimensional result shows that the hypersurface condition forces the root to be maximal. Here and below a hypersurface means that the embedding dimension is one more than the Krull dimension.

Theorem 1.2 (Finiteness and rigidity). *Fix $n \geq 4$ and $\delta \geq 1$. Up to permutation, only finitely many signatures $\mathbf{p}$ satisfying (1.1) admit a root $\delta\vec{\tau} = \vec{\omega}$ for which $R(\mathbf{p}, \vec{\tau})$ is a hypersurface. Every such signature satisfies*

$$(1.9) \quad N\nu = \delta, \quad \text{equivalently} \quad \sum_{i=1}^n \frac{1}{p_i} + \frac{\delta}{N} = 1.$$

Consequently $\delta = d_{\max}$ and $\vec{\tau} = \vec{\omega}$.

The rigidity equation does more than reduce the classification to a finite search. It identifies the root ring with a quotient of an affine semigroup ring determined entirely by the pairwise gcds of the p_i . The general invariant-theoretic classification used here is Watanabe's: finite abelian diagonal subgroups of $\mathrm{SL}_n(\mathbb{C})$ with complete-intersection invariant rings are encoded by special data [11, Definition 1.1, Proposition 1.7, and Theorem 2.1]. Our contribution is the arithmetic specialization to the particular group $G_{\mathbf{p}}$: the special-datum condition becomes a multiplicative-ultrametric condition on $d_{ij} = \gcd(p_i, p_j)$, the datum is reconstructed canonically from the gcd matrix, and the resulting root-ring presentation, embedding dimension, and primitive weights are explicit. The associated laminar gcd forest gives the Hilbert basis and one toric binomial for each internal node. Corollary 5.4 recovers the full higher-dimensional hypersurface classification as the zero- or one-node case. The two-level signature $(2, 4, 8, 11)$ shows that the complete-intersection theorem is genuinely stronger than a single-common-gcd construction.

The isolated locus has a particularly clean arithmetic description: it is exactly the pairwise-coprime locus, and the root index is the corresponding Egyptian-fraction defect. The Euler product below is the classical density of pairwise-coprime tuples [9], and its product-height mean is a standard Selberg–Delange calculation [1, 8]. What is new here is the transfer from ordinary product height to the root-index defect, including control of the Egyptian-fraction cusps and the exact parity split. Counting all such signatures, rather than a chosen explicit subfamily, gives the mean asymptotic

$$\sum_{\delta \leq X} I_n^{\mathrm{iso}}(\delta) \sim \frac{\mathfrak{S}_n}{n!(n-1)!} X(\log X)^{n-1}, \quad \mathfrak{S}_n = \prod_{\ell} \left(1 - \frac{1}{\ell}\right)^{n-1} \left(1 + \frac{n-1}{\ell}\right).$$

Theorem 6.2 also determines the contribution of each parity class and gives logarithmic-power lower bounds at individual indices.

Section 2 proves the root criterion. Section 3 establishes the diagonal-invariant description and the two numerical identities. Section 4 proves rigidity, and Section 5 classifies complete intersections and hypersurfaces. Section 6 treats the isolated locus and its mean asymptotic.

2. THE GRADING GROUP AND ROOTS OF THE DUALIZING ELEMENT

We use the degree homomorphism

$$(2.1) \quad \deg : L \longrightarrow \mathbb{Q}, \quad \deg \vec{c} = 1, \quad \deg \vec{x}_i = \frac{1}{p_i}.$$

Its image is $\frac{1}{N}\mathbb{Z}$. For every $\vec{v} \in L$ one has

$$(2.2) \quad N\vec{v} = N \deg(\vec{v})\vec{c}.$$

It follows that $\ker(\deg) = \mathrm{tors}(L)$ and that its exponent divides N . Thus

$$(2.3) \quad 0 \longrightarrow \mathrm{tors}(L) \longrightarrow L \xrightarrow{\deg} \frac{1}{N}\mathbb{Z} \longrightarrow 0$$

is exact. Notice also that

$$(2.4) \quad L/\mathbb{Z}\vec{c} \cong \bigoplus_{i=1}^n \mathbb{Z}/p_i\mathbb{Z}, \quad \vec{\omega} \equiv - \sum_i \vec{x}_i \pmod{\mathbb{Z}\vec{c}}.$$

Lemma 2.1. *If e is coprime to N , multiplication by e on L is injective.*

Proof. An element killed by e has degree zero, hence belongs to $\mathrm{tors}(L)$. It is also killed by N by (2.2). Since $\gcd(e, N) = 1$, it is zero. $\square$

Proof of Theorem 1.1. Suppose first that $\delta\vec{\tau} = \vec{\omega}$. Reducing modulo $\mathbb{Z}\vec{c}$ in (2.4) shows that multiplication by δ must be invertible in every $\mathbb{Z}/p_i\mathbb{Z}$. Hence

$$(2.5) \quad \gcd(\delta, N) = 1.$$

Taking degrees and multiplying by N gives

$$k = N \deg \vec{\omega} = \delta N \deg \vec{\tau},$$

so $\delta \mid k$. Therefore $\delta \mid d_{\max}$.

It remains to construct the maximal root. Choose integers b_i satisfying $b_i p_i \equiv 1 \pmod{d_{\max}}$. Since $d_{\max} \mid k$ and $\gcd(d_{\max}, N) = 1$,

$$N \left(1 - \sum_i b_i \right) \equiv N - \sum_i \frac{N}{p_i} = k \equiv 0 \pmod{d_{\max}},$$

and hence $1 - \sum_i b_i$ is divisible by $d_{\max}$. The element

$$(2.6) \quad \vec{\omega} = \frac{1 - \sum_i b_i}{d_{\max}} \vec{c} + \sum_i \frac{b_i p_i - 1}{d_{\max}} \vec{x}_i$$

belongs to L and satisfies $d_{\max} \vec{\omega} = \vec{\omega}$.

If $\delta \mid d_{\max}$, then $\vec{\tau} = (d_{\max}/\delta) \vec{\omega}$ is a δ -th root. Conversely, any root has $\delta \mid d_{\max}$ by the first paragraph, and Lemma 2.1 applied to

$$\delta \left(\vec{\tau} - \frac{d_{\max}}{\delta} \vec{\omega} \right) = 0$$

proves uniqueness. □

Remark 2.2. Write a root in normal form modulo $\mathbb{Z}\vec{c}$ as $\vec{\tau} \equiv \sum_i s_i \vec{x}_i$. Equation (1.5) gives

$$(2.7) \quad \delta s_i \equiv -1 \pmod{p_i}.$$

In particular the character of $\vec{\tau}$ at the i -th stacky point is faithful. The minus sign in (2.7) will be important in Section 4.

3. INVARIANT RINGS, THE CANONICAL SHIFT, AND MULTIPLICITY

Fix a root $\delta\vec{\tau} = \vec{\omega}$ and put

$$(3.1) \quad \mu = \deg \vec{\tau} = \frac{\nu}{\delta}, \quad u = N\mu = \frac{k}{\delta} \in \mathbb{Z}_{>0}.$$

The quotient $L/\mathbb{Z}\vec{\tau}$ is finite. Its Cartier dual

$$(3.2) \quad G_{\vec{\tau}} := \text{Hom}(L/\mathbb{Z}\vec{\tau}, \mathbb{C}^*)$$

acts diagonally on T . We emphasize that $L/\mathbb{Z}\vec{\tau}$ is the character group of $G_{\vec{\tau}}$, not canonically the group $G_{\vec{\tau}}$ itself.

Proposition 3.1. *The group and its order are*

$$(3.3) \quad G_{\vec{\tau}} = \left\{ \text{diag}(\alpha_1, \dots, \alpha_n) : \alpha_i^{p_i} = \rho := \prod_{j=1}^n \alpha_j, \quad \rho^u = 1 \right\},$$

$$(3.4) \quad |G_{\vec{\tau}}| = \mu \prod_{i=1}^n p_i.$$

Moreover,

$$(3.5) \quad R(\mathbf{p}, \vec{\tau}) = T^{G_{\vec{\tau}}}.$$

Proof. A character annihilating $\vec{\tau}$ annihilates $\vec{\omega} = \delta\vec{\tau}$. The relations in L and the identity $\alpha(\vec{\omega}) = 1$ give $\alpha_i^{p_i} = \alpha(\vec{c}) = \prod_j \alpha_j = \rho$. Since $N\vec{\tau} = u\vec{c}$, one also has $\rho^u = 1$.

Conversely, the displayed equations imply $\alpha(\vec{\omega}) = 1$ and $\alpha(N\vec{\tau}) = 1$. Thus $\alpha(\vec{\tau})^\delta = \alpha(\vec{\tau})^N = 1$; by (2.5), $\alpha(\vec{\tau}) = 1$. This proves (3.3).

The torsion group in (2.3) has order $\prod_i p_i/N$, while the image of $\vec{\tau}$ has index u in $\frac{1}{N}\mathbb{Z}$. Therefore

$$|L/\mathbb{Z}\vec{\tau}| = u \frac{\prod_i p_i}{N} = \mu \prod_i p_i,$$

which proves (3.4). Finally, a homogeneous element of T is fixed by all characters of $L/\mathbb{Z}\vec{\tau}$ precisely when its L -degree belongs to $\mathbb{Z}\vec{\tau}$. Its nonnegative-degree part is (3.5). $\square$

The Brieskorn–Pham hypersurface T is normal: its only singular point is the origin and $\dim T \geq 2$. Since finite-group invariants preserve normality, $R(\mathbf{p}, \vec{\tau})$ is finitely generated and normal. Give T the rational $\vec{\tau}$ -degree

$$(3.6) \quad \deg_{\vec{\tau}} X_i = \frac{1}{\mu p_i}, \quad \deg_{\vec{\tau}} \left(\sum_i X_i^{p_i} \right) = \frac{1}{\mu}.$$

The invariant part has integral degrees. Molien’s formula gives

$$(3.7) \quad P_R(t) = \frac{1}{|G_{\vec{\tau}}|} \sum_{g \in G_{\vec{\tau}}} \frac{1 - \rho(g)t^{1/\mu}}{\prod_{i=1}^n (1 - \alpha_i(g)t^{1/(\mu p_i)}}.$$

Proposition 3.2. *If $R = R(\mathbf{p}, \vec{\tau})$ has a hypersurface presentation*

$$R \cong \mathbb{C}[z_1, \dots, z_n]/(f), \quad \deg z_i = a_i, \quad \deg f = h,$$

then

$$(3.8) \quad h - \sum_i a_i = \delta, \quad \frac{h}{\prod_i a_i} = \mu^{n-2}.$$

With the convention $M(s)_m = M_{m+s}$, the hypersurface canonical-module formula also gives $K_R \cong R(\delta)$.

Proof. Replace t by t^{-1} in (3.7), factor the powers of t , and reindex the sum by $g \mapsto g^{-1}$. Since

$$\frac{1}{\mu} - \sum_i \frac{1}{\mu p_i} = \frac{\nu}{\mu} = \delta, \quad \rho(g) = \prod_i \alpha_i(g),$$

we obtain

$$(3.9) \quad P_R(t^{-1}) = (-1)^{n-1} t^{-\delta} P_R(t).$$

For the hypersurface Hilbert series

$$P_R(t) = \frac{1 - t^h}{\prod_i (1 - t^{a_i})},$$

the same calculation gives the exponent $-(h - \sum_i a_i)$. This proves the first identity.

For the second identity, consider the pole at $t = 1$. A nonidentity element of $G_{\vec{\tau}}$ cannot fix $n - 1$ coordinates: if all but α_i are 1, then $\rho = 1$ from any fixed coordinate and hence $\alpha_i = 1$ as well. Thus only the identity contributes to the pole of order $n - 1$. Its contribution before division by $|G_{\vec{\tau}}|$ has leading coefficient

$$\frac{1/\mu}{\prod_i (1/(\mu p_i))} = \mu^{n-1} \prod_i p_i.$$

Using (3.4), the leading coefficient of $P_R(t)$ is μ^{n-2} . The hypersurface expression has leading coefficient $h/\prod_i a_i$, and the result follows. $\square$

4. HYPERSURFACE RIGIDITY IN FOUR OR MORE VARIABLES

Assume throughout this section that $n \geq 4$. Set

$$(4.1) \quad N_i = \text{lcm}\{p_j : j \neq i\}, \quad q_i = \mu p_i N_i = u \gcd(p_i, N_i) \in \mathbb{Z}_{>0}.$$

Let $\bar{T} = \mathbb{C}[\bar{X}_1, \dots, \bar{X}_n]$ with its L -grading and let $T = \bar{T}/(\bar{F})$, where $\bar{F} = \sum_i \bar{X}_i^{p_i}$. Put

$$\bar{R}_m = \bar{T}_{m\vec{\tau}}, \quad \bar{R} = \bigoplus_{m \geq 0} \bar{R}_m,$$

denote the natural map $\bar{R} \rightarrow R$ by ϕ , and write $Y_i = \bar{X}_i^{p_i}$.

Lemma 4.1. *For every i one has*

$$(4.2) \quad N_i \vec{\tau} = q_i \vec{x}_i.$$

A pure power $\bar{X}_i^m$ has degree in $\mathbb{Z}\vec{\tau}$ if and only if $q_i \mid m$.

Proof. Multiplication of (4.2) by δ gives

$$\delta N_i \vec{\tau} = N_i \vec{\omega} = N_i \left(\nu + \frac{1}{p_i} \right) \vec{c} - N_i \vec{x}_i = \nu p_i N_i \vec{x}_i = \delta q_i \vec{x}_i.$$

Multiplication by δ is injective by Lemma 2.1, proving (4.2).

If $m \vec{x}_i = r \vec{\tau}$, reduction modulo $\mathbb{Z}\vec{c}$ in every component $j \neq i$ and (2.7) show that $p_j \mid r$. Hence $N_i \mid r$, and writing $r = a N_i$ in (4.2) gives $m \vec{x}_i = a q_i \vec{x}_i$. Applying the degree homomorphism yields $m = a q_i$, and hence $q_i \mid m$. The converse follows immediately from (4.2). $\square$

We now give the root version of the generator argument in [5, Lemmas 2.1–2.8 and Proposition 2.9]. The details are included because both the pure-power exponents and one residue congruence change.

Every $\vec{v} \in L$ has a unique normal form

$$(4.3) \quad \vec{v} = \ell(\vec{v}) \vec{c} + \sum_i a_i(\vec{v}) \vec{x}_i, \quad 0 \leq a_i(\vec{v}) < p_i,$$

and we put $M(\vec{v}) = \prod_i \bar{X}_i^{a_i(\vec{v})}$. Thus every element of $\bar{T}_{\vec{v}}$ is $M(\vec{v})$ times a polynomial of ordinary degree $\ell(\vec{v})$ in the Y_i .

Let $\mathfrak{m} = \bigoplus_{m \geq 0} \bar{R}_m$. Since a hypersurface R has embedding dimension n , choose monomial lifts $\bar{\Xi} = \{\bar{z}_1, \dots, \bar{z}_n\} \subset \bar{R}$ whose images form a basis of $\mathfrak{m}/\mathfrak{m}^2$. Such lifts exist because invariant monomials span every homogeneous piece of R , so their classes span $\mathfrak{m}/\mathfrak{m}^2$ and a basis may be extracted from them. They generate R : if B is the subalgebra generated by their images, then $\mathfrak{m} = B_+ + \mathfrak{m}^2$, and induction on the positive integer degree gives $R = B$.

Lemma 4.2 (Homogeneous division). *For $m \geq 0$ and $Z \in \bar{R}_m$, there are $P \in \mathbb{C}[Y_1, \dots, Y_n]_{\ell(m\vec{\tau})-1}$ and a homogeneous $Q \in \mathbb{C}[\bar{\Xi}] \cap \bar{R}_m$ such that*

$$(4.4) \quad Z = M(m\vec{\tau}) P \bar{F} + Q.$$

Proof. Choose Q with $\phi(Q) = \phi(Z)$. Homogeneity gives $Z - Q = P' \bar{F}$ for $P' \in \bar{T}_{m\vec{\tau} - \vec{c}}$. Normal form (4.3) gives $M(m\vec{\tau} - \vec{c}) = M(m\vec{\tau})$ and $\ell(m\vec{\tau} - \vec{c}) = \ell(m\vec{\tau}) - 1$. Hence $P' = M(m\vec{\tau}) P$ for some $P \in \mathbb{C}[Y_1, \dots, Y_n]_{\ell(m\vec{\tau})-1}$; substituting this into $Z - Q = P' \bar{F}$ proves the assertion. $\square$

Lemma 4.3 (Root monomial-generator lemma). *Let*

$$I = \{i : \bar{X}_i^{q_i} \in \bar{\Xi}\}.$$

Then $|I| \geq n - 1$. If $I = \{1, \dots, n\}$, then $q_i = 1$ for every i and $R = T$.

Proof. We first establish two projection facts. Suppose $i \neq j$, $i \notin I$, and no generator is of the form $\overline{X}_i^a \overline{X}_j^b$ with $a, b > 0$. Apply Lemma 4.2 to $Z = \overline{X}_i^{q_i}$ and set every variable except $\overline{X}_i, \overline{X}_j$ equal to zero. The projected Q can involve only pure generators in $\overline{X}_j$: a pure generator in $\overline{X}_i$ would necessarily be $\overline{X}_i^{q_i}$, contrary to $i \notin I$, and mixed generators were excluded. It cannot vanish, since otherwise the monomial $\overline{X}_i^{q_i}$ would be a multiple of $Y_i + Y_j$. The pure-power criterion and minimality show that any surviving pure generator is $\overline{X}_j^{q_j}$. Moreover, divisibility of $\overline{X}_i^{q_i} - Q$ by $M(q_i \vec{x}_i)$ forces $M(q_i \vec{x}_i) = 1$, and therefore

$$(4.5) \quad p_i \mid q_i.$$

The nonzero projected Q also shows that $j \in I$. We have therefore proved the first projection fact in the following form: if $i \notin I$ and no mixed generator supported on $\{i, j\}$ occurs, then $j \in I$ and $p_i \mid q_i$.

Next let i, j, k be distinct, with $i, j \notin I$ and $k \in I$. Suppose no generator is a monomial supported on $\{i, j, k\}$ of the form $\overline{X}_i^a \overline{X}_j^b \overline{X}_k^c$ with $(a, b) \neq (0, 0)$ and $c > 0$. Applying the first fact to the pair (i, k) gives $p_i \mid q_i$. Write $q = q_i/p_i$, apply Lemma 4.2 to Y_i^q , and project to these three variables. With the leading coefficient normalized, write

$$P = Y_i^{q-1} + A_1 Y_i^{q-2} + \cdots + A_{q-1}, \quad A_l \in \mathbb{C}[Y_j, Y_k].$$

Every monomial of the projected Q is either divisible by $\overline{X}_i \overline{X}_j$ or is a pure power of $\overline{X}_k$. Since the other terms in the projected division identity belong to $\mathbb{C}[Y_i, Y_j, Y_k]$, so does the projected Q ; consequently

$$(4.6) \quad Q \in (Y_i Y_j) \mathbb{C}[Y_i, Y_j, Y_k] + \mathbb{C}[Y_k].$$

If $q = 1$, the identity $Y_i = Y_i + Y_j + Y_k + Q$ forces a nonzero Y_j term in Q , contradicting (4.6). Thus $q > 1$. In

$$Y_i^q = P(Y_i + Y_j + Y_k) + Q,$$

the constant coefficient in Y_i says that $A_{q-1}(Y_j + Y_k)$ belongs to $\mathbb{C}[Y_k]$. Hence $A_{q-1} = 0$ (and the constant coefficient of Q is zero). Moving upwards through the coefficients of Y_i and reducing each identity modulo Y_j now gives $Y_j \mid A_l$ for every l . Finally, the coefficient of Y_i^{q-1} , reduced modulo Y_j , contains the uncanceled term Y_k . This contradicts (4.6).

Hence a generator $\overline{X}_i^a \overline{X}_j^b \overline{X}_k^c$ with $(a, b) \neq (0, 0)$ and $c > 0$ must occur.

Put $s = n - |I|$. If $s \geq 2$, the first projection fact supplies at least $\binom{s}{2}$ generators joining pairs of missing indices, while the second supplies at least $|I|$ additional generators joining them to the indices in I . Together with the $|I|$ pure generators,

$$n = |\Xi| \geq 2(n - s) + \frac{s(s - 1)}{2}.$$

Equivalently $n \leq -s(s - 5)/2$, whose right-hand side is at most 3 for $0 \leq s \leq n$. This contradicts $n \geq 4$, so $|I| \geq n - 1$.

Finally assume that every pure power occurs. Write $\vec{r} \equiv \sum_i s_i \vec{x}_i \pmod{\mathbb{Z}\vec{c}}$ and choose a sufficiently large integer r with

$$(4.7) \quad r \equiv -\delta \pmod{N}.$$

Equation (2.7) gives $a_i(r\vec{r}) \equiv r s_i \equiv 1 \pmod{p_i}$, so there is a monomial $\overline{X}_i G \in \overline{R}_r$ with $\overline{X}_i \nmid G$: after increasing r within its congruence class, the normal-form polynomial degree is positive and may be placed in variables other than Y_i . If $q_i > 1$, a polynomial in the pure generators $\overline{X}_i^{q_i}$ has no term of degree one in $\overline{X}_i$. Comparing such terms in (4.4) gives

$$\overline{X}_i G = M(r\vec{r}) P|_{\overline{X}_i=0} \sum_{l \neq i} Y_l,$$

which is impossible: $\sum_{l \neq i} Y_l$ has at least two relatively prime monomial terms and therefore cannot divide a monomial. Thus every $q_i = 1$, and the images of the variables generate R ; hence $R = T$. $\square$

Proposition 4.4 (Root rigidity). *If $n \geq 4$ and $R(\mathbf{p}, \vec{\tau})$ is a hypersurface, then*

$$(4.8) \quad u = N \deg \vec{\tau} = 1.$$

Equivalently, $N\nu = \delta$.

Proof. If $R = T$, each X_i belongs to R , so Lemma 4.1 gives $q_i \mid 1$. Hence $1 = q_i = u \gcd(p_i, N_i)$ and $u = 1$.

Otherwise Lemma 4.3 gives a unique missing index i . The other generators are $\overline{X}_l^{q_l}$ for $l \neq i$. The remaining generator involves some variable $\overline{X}_m$ with $m \neq i$; otherwise the pure-power criterion and minimality would put i in I . Choose distinct $j, k \notin \{i, m\}$, possible since $n \geq 4$.

There is no two-variable generator $\overline{X}_i^a \overline{X}_j^b$. The first projection argument in the proof of Lemma 4.3 therefore gives $p_i \mid q_i$. Put $q = q_i/p_i$ and apply Lemma 4.2 to Y_i^q . Project to $\mathbb{C}[\overline{X}_i, \overline{X}_j, \overline{X}_k]$. The remaining generator vanishes under this projection because it contains $\overline{X}_m$, and consequently

$$\pi(Q) \in \mathbb{C}[\overline{X}_j^{q_j}, \overline{X}_k^{q_k}].$$

Writing

$$\pi(P) = Y_i^{q-1} + A_1 Y_i^{q-2} + \cdots + A_{q-1}, \quad A_l \in \mathbb{C}[Y_j, Y_k],$$

and comparing powers of Y_i in $Y_i^q = \pi(P)(Y_i + Y_j + Y_k) + \pi(Q)$ gives

$$A_l = (-Y_j - Y_k)^l, \quad \pi(Q) = (-Y_j - Y_k)^q.$$

Since the latter polynomial belongs to $\mathbb{C}[\overline{X}_j^{q_j}, \overline{X}_k^{q_k}]$, comparison of its monomials (the end terms when $q = 1$, and the terms with exponents 1 and $q - 1$ when $q > 1$) yields

$$(4.9) \quad q_j \mid p_j, \quad q_k \mid p_k.$$

Now (4.1) gives

$$(4.10) \quad \frac{p_j}{q_j} = \frac{N}{uN_j}, \quad \frac{p_k}{q_k} = \frac{N}{uN_k}$$

are integers. Thus u divides both N/N_j and N/N_k . These two integers are coprime: a prime divides N/N_j only when its maximal exponent among the p_ℓ is attained uniquely at j , and the same prime cannot be attained uniquely at the distinct index k . Hence $u = 1$. $\square$

Proof of Theorem 1.2 for $n \geq 4$. Proposition 4.4 gives $\nu = \delta/N$, hence (1.9). In particular $k = N\nu = \delta$; together with $\gcd(\delta, N) = 1$, this says $d_{\max} = \delta$.

For finiteness, order the signature so that $p_1 \leq \cdots \leq p_n$. Once $p_1, \dots, p_{r-1}$ are fixed, put

$$c_r = 1 - \sum_{i < r} \frac{1}{p_i} > 0.$$

Equation (1.9) and $p_r \leq p_i, N$ for $i \geq r$ give

$$c_r = \sum_{i=r}^n \frac{1}{p_i} + \frac{\delta}{N} \leq \frac{n - r + 1 + \delta}{p_r}.$$

Thus $p_r \leq (n - r + 1 + \delta)/c_r$. Induction on r leaves only finitely many ordered signatures. $\square$

5. COMPLETE INTERSECTIONS AND THE GCD FOREST

For the rest of this section, assume the rigid maximal-root conditions

$$(5.1) \quad \delta = N \left(1 - \sum_{i=1}^n \frac{1}{p_i} \right) > 0, \quad \gcd(\delta, N) = 1.$$

Theorem 1.1 gives the unique maximal root $\vec{\tau}$, and $N\vec{\tau} = \vec{c}$. Write $\vec{\tau} \equiv \sum_i s_i \vec{x}_i \pmod{\mathbb{Z}\vec{c}}$. The root equation gives $\delta s_i \equiv -1 \pmod{p_i}$. Since δ is a unit modulo every p_i , a monomial X^e has degree in $\mathbb{Z}\vec{\tau}$ precisely when its exponents have a common residue modulo the p_i . Thus, with

$$(5.2) \quad S(\mathbf{p}) = \{e \in \mathbb{N}^n : e_i \equiv t \pmod{p_i} \text{ for some } t \in \mathbb{Z}\},$$

exactness of finite-group invariants gives

$$(5.3) \quad R(\mathbf{p}, \vec{\tau}) \cong \mathbb{C}[S(\mathbf{p})]/(F), \quad F = \sum_i X_i^{p_i}.$$

Indeed, cancellation by the nonzero invariant F shows that $(F) \cap \mathbb{C}[S] = F\mathbb{C}[S]$.

For a signature $\mathbf{p}$, let $\Gamma(\mathbf{p})$ be the graph on $\{1, \dots, n\}$ in which i and j are adjacent exactly when $\gcd(p_i, p_j) > 1$. The right combinatorial refinement of this gcd graph is a laminar forest. Put

$$d_{ij} = \gcd(p_i, p_j), \quad q_i = \gcd(p_i, \text{lcm}_{j \neq i} p_j).$$

Prime by prime one has

$$(5.4) \quad q_i = \text{lcm}_{j \neq i} d_{ij}, \quad \gcd(q_i, q_j) = d_{ij}.$$

We say that the pair-gcd matrix is a *multiplicative ultrametric* if, for every three distinct indices i, j, k , after possibly interchanging them one has

$$(5.5) \quad d_{ij} = d_{ik} \mid d_{jk}.$$

Thus two of the three pairwise gcds must be equal, and their common value must divide the third.

There is a canonical recursive way to read (5.5). Start with the connected components of the gcd graph. A singleton is a leaf of weight one. For a nontrivial component C , start with $w_C = 1$, set

$$m_C = \gcd_{i \in C} \frac{q_i}{w_C},$$

and normalize the pair gcds by $w_C m_C$. The graph on C in which i, j are joined when $d_{ij}/(w_C m_C) > 1$ has as its connected components the children of C . A singleton child $\{i\}$ is accepted precisely when $q_i = w_C m_C$; give it that weight. For every nontrivial child C' , put $w_{C'} = w_C m_C$ and repeat. The process is said to *succeed* if every internal multiplier m_C is greater than one and every terminal singleton is accepted. When it succeeds, the resulting subsets are nested or disjoint. We denote the resulting laminar forest, including its singleton leaves, by $\mathcal{D}$. Its internal nodes are denoted by $\mathcal{D}^\circ$, and

$$t(\mathbf{p}) = |\mathcal{D}^\circ|.$$

For $J \in \mathcal{D}^\circ$, its children have a common weight $w_J m_J$; we write $K \prec J$ when K is one of these children.

Lemma 5.1 (Ultrametries and the gcd forest). *The pair-gcd matrix satisfies (5.5) if and only if the recursive gcd-forest construction succeeds.*

Proof. We argue recursively with the invariant that the current weight w divides d_{ij} for all leaves i, j in the current component C ; this is clear for the initial weight $w = 1$, and the argument below shows that it is preserved for every child. Put

$$d'_{ij} = d_{ij}/w \quad (i, j \in C), \quad q'_i = q_i/w \quad (i \in C).$$

The identities in (5.4) persist in the form

$$q'_i = \text{lcm}_{j \in C \setminus \{i\}} d'_{ij}, \quad \gcd(q'_i, q'_j) = d'_{ij}.$$

Indeed, at the split that created C , every gcd between a leaf of C and a leaf outside C was a divisor of w . On the other hand, since C is nontrivial, it contains a second leaf whose gcd with i is a multiple of w . Thus the outside terms in $q_i = \text{lcm}_{j \neq i} d_{ij}$ divide an inside term and disappear after division by w . The gcd identity follows directly from (5.4). If $d'_{ij} > 1$ and $d'_{jk} > 1$, the ultrametric condition forces $d'_{ik} > 1$; hence adjacency is transitive and the new connected components are disjoint cliques. Choose $a, b \in C$ for which d'_{ab} is minimal. Applied to the triple a, b, i , the ultrametric condition shows that d'_{ab} divides both d'_{ai} and d'_{bi} , and one of the latter two equals d'_{ab} . A second application to a, i, j shows that d'_{ab} divides every d'_{ij} . Since $\gcd(q'_a, q'_b) = d'_{ab}$, it follows that

$$m_C = \gcd_{i \in C} q'_i = d'_{ab} > 1.$$

Thus m_C is exactly the branching weight at this stage. Division by m_C again produces a multiplicative ultrametric. If a child is the singleton $\{i\}$, then

$$q'_i/m_C = \text{lcm}_{j \in C \setminus \{i\}} (d'_{ij}/m_C) = 1,$$

so the singleton is accepted. Every nontrivial child is handled inductively. At each step the positive integers q'_i strictly decrease, so the process terminates and succeeds.

Conversely, suppose the forest succeeds. For two leaves, their gcd is the common weight of the two branches immediately below their least common ancestor: at every split, distinct children have normalized gcd one, by (5.4). Among three leaves, two pairs have the same least branching weight, and this weight divides the third pair's weight. This is precisely (5.5). $\square$

We shall use the following elementary consequence of Watanabe's definition [11, Definitions 1.1 and 1.3 and Proposition 1.7]. Recall that a special datum is a laminar family $\mathcal{E}$ containing all singletons, with weight one on every maximal node and with all immediate children of a node carrying the same proper multiple of the parent's weight. Its diagonal group $G_{\mathcal{E}}$ is generated by

$$(5.6) \quad (1, \dots, \zeta, \dots, \zeta^{-1}, \dots, 1),$$

where the two nontrivial coordinates lie in distinct children K, K' of one node and $\zeta \in \mu_{w_K} = \mu_{w_{K'}}$.

Lemma 5.2 (Pair subgroups of a special datum). *For distinct leaves i, j , let $b_{ij} = 1$ if they lie in different maximal nodes. Otherwise let $J(i, j)$ be the least node containing both and let $K_i, K_j \prec J(i, j)$ be the children containing them; put*

$$b_{ij} = w_{K_i} = w_{K_j}.$$

Then the subgroup of $G_{\mathcal{E}}$ supported on the two coordinates i, j is exactly

$$H_{ij}(G_{\mathcal{E}}) = \{(1, \dots, \zeta, \dots, \zeta^{-1}, \dots, 1) : \zeta \in \mu_{b_{ij}}\}.$$

Moreover $G_{\mathcal{E}}$ is generated by these pair-supported subgroups. In particular the matrix (b_{ij}) is a multiplicative ultrametric.

Proof. The monomial

$$M_K = \left(\prod_{l \in K} X_l \right)^{w_K}$$

attached to every node is $G_{\mathcal{E}}$ -invariant by the defining generators (5.6). Indeed, such a pair generator meets K in zero or two selected leaves, or in one leaf; in the last case its order divides w_K , since descendant weights are multiples of ancestor-child weights.

Suppose that an element is supported on i, j . All defining generators have determinant one, so its two possibly nontrivial entries are α, α^{-1} . If i, j lie in different maximal nodes, invariance of the weight-one monomial belonging to the maximal node containing i gives $\alpha = 1$. Otherwise invariance of M_{K_i} gives $\alpha^{b_{ij}} = 1$. Conversely, (5.6) with the two leaves i, j realizes every $\alpha \in \mu_{b_{ij}}$. This proves the formula for H_{ij} . The assertion that these subgroups generate is precisely the generating rule (5.6). Finally, among three leaves, two pairs branch at the same node and have the same b -value, while that value divides the third pair's (possibly deeper) branching weight. This is the multiplicative-ultrametric condition. $\square$

Theorem 5.3 (Complete-intersection gcd-forest classification). *Let $n \geq 4$, assume (5.1), and let $R = R(\mathbf{p}, \vec{\tau})$ be the corresponding maximal canonical root ring. The following conditions are equivalent.*

- (1) R is a complete intersection.
- (2) The invariant semigroup ring $\mathbb{C}[S(\mathbf{p})]$ is a complete intersection.
- (3) The pair-gcd matrix satisfies (5.5).
- (4) The recursive gcd-forest construction succeeds.

When these conditions hold, put

$$M_J = \left(\prod_{i \in J} X_i \right)^{w_J} \quad (J \in \mathcal{D}), \quad Y_i = M_{\{i\}} = X_i^{q_i}, \quad c_i = \frac{p_i}{q_i}.$$

Then the exponent vectors $w_J \mathbf{1}_J$, $J \in \mathcal{D}$, are the Hilbert basis of $S(\mathbf{p})$, and there are presentations

$$(5.7) \quad \mathbb{C}[S(\mathbf{p})] \cong \mathbb{C}[(M_J)_{J \in \mathcal{D}}] / \left(M_J^{m_J} - \prod_{K \prec J} M_K : J \in \mathcal{D}^\circ \right),$$

$$(5.8) \quad R \cong \mathbb{C}[(M_J)_{J \in \mathcal{D}}] / \left(\left(M_J^{m_J} - \prod_{K \prec J} M_K \right)_{J \in \mathcal{D}^\circ}, \sum_{i=1}^n Y_i^{c_i} \right).$$

The $t(\mathbf{p})$ toric binomials in (5.7) form a regular sequence, and after adjoining the last equation in (5.8) one still has a regular sequence. If

$$\epsilon = [\text{there is an } i \text{ with } q_i = p_i] = [\text{there is an } i \text{ with } c_i = 1],$$

then

$$(5.9) \quad \text{edim } R = n + t(\mathbf{p}) - \epsilon, \quad \text{codim } R = t(\mathbf{p}) + 1 - \epsilon.$$

In particular, this is a necessary and sufficient arithmetic classification of all complete intersections among the rigid maximal-root rings in every number $n \geq 4$ of variables.

The presentation is weighted homogeneous for

$$(5.10) \quad \begin{aligned} \deg M_J &= w_J \sum_{i \in J} \frac{N}{p_i}, \\ \deg \left(M_J^{m_J} - \prod_{K \prec J} M_K \right) &= w_J m_J \sum_{i \in J} \frac{N}{p_i}, \\ \deg \sum_i Y_i^{c_i} &= N. \end{aligned}$$

These weights are primitive, and their complete-intersection Gorenstein parameter is δ .

Proof. We give the semigroup reduction first. The generalized Chinese remainder theorem says that an exponent vector $e \in \mathbb{N}^n$ has a common residue modulo the p_i if and only if

$$e_i \equiv e_j \pmod{d_{ij}} \quad (i, j = 1, \dots, n).$$

Consequently $S(\mathbf{p})$ depends only on the pairwise gcds, equivalently on the pure periods q_i by (5.4). As in (5.3),

$$R \cong A/(F), \quad A = \mathbb{C}[S(\mathbf{p})], \quad F = \sum_i X_i^{p_i}.$$

The ring A is a domain, so F is a nonzerodivisor. The regular-element deformation theorem for complete-intersection local rings [7, Theorem 21.1], applied at the homogeneous maximal ideal, gives

$$A \text{ is a complete intersection} \iff A/(F) \text{ is a complete intersection.}$$

For completeness, this local statement gives the graded affine statement used here as follows. In a minimal homogeneous polynomial presentation $P \twoheadrightarrow A$, the minimal number of homogeneous generators of the kernel I is $\dim_{\mathbb{C}} I/\mathfrak{m}_P I$, and localization at the homogeneous maximal ideal preserves this number. Thus $A_{\mathfrak{m}}$ is a complete intersection if and only if I is generated by $\text{ht } I$ homogeneous elements; because P is Cohen–Macaulay, these form a regular sequence. The same argument applies to $A/(F)$. This proves the equivalence of the first two conditions.

We next apply the abelian invariant-ring classification. Set

$$G_{\mathbf{p}} = \left\{ (\xi_1, \dots, \xi_n) \in \prod_i \mu_{p_i} : \prod_i \xi_i = 1 \right\}.$$

It is a finite abelian subgroup of $\text{SL}(n, \mathbb{C})$, it has no pseudoreflections, and $A = \mathbb{C}[X_1, \dots, X_n]^{G_{\mathbf{p}}}$. Moreover

$$(5.11) \quad G_{\mathbf{p}} = \left\langle (1, \dots, \zeta, \dots, \zeta^{-1}, \dots, 1) : \zeta \in \mu_{d_{ij}}, \ i < j \right\rangle.$$

For fixed i, j , the subgroup acting trivially off those two coordinates is exactly the displayed copy of $\mu_{d_{ij}}$: a pair-supported element has entries ζ, ζ^{-1} , with $\zeta^{p_i} = \zeta^{p_j} = 1$. To prove that these pair subgroups generate $G_{\mathbf{p}}$, argue prime by prime. For a prime ℓ , choose an index at which $v_{\ell}(p_i)$ is maximal. Pairing that coordinate with each of the others generates a subgroup of order $\ell^{\sum_i v_{\ell}(p_i) - \max_i v_{\ell}(p_i)}$, which is exactly the ℓ -part of $|G_{\mathbf{p}}|$.

Watanabe’s necessity theorem for invariant complete intersections of finite abelian groups [11, Theorem 2.1], together with the converse construction [11, Proposition 1.7], is stated in a fixed simultaneous eigenbasis. Since $G_{\mathbf{p}} \subset \text{SL}(n, \mathbb{C})$, it says that A is a complete intersection precisely when $G_{\mathbf{p}}$ comes from a special datum. For such a datum the associated invariant ring is

$$(5.12) \quad \mathbb{C} \left[\left(\prod_{i \in J} X_i \right)^{w_J} : J \in \mathcal{E} \right].$$

If A is a complete intersection, the pair-supported subgroup of $G_{\mathbf{p}}$ on coordinates i, j is $\mu_{d_{ij}}$, by (5.11). Lemma 5.2 therefore identifies d_{ij} with the branching weight b_{ij} of the special datum. The last assertion of that lemma gives (5.5). This proves the implication from the second condition to the third.

Lemma 5.1 proves the equivalence of the third and fourth conditions and shows that the successful recursion produces the canonical laminar forest determined by the pair-gcd matrix.

Conversely, give the resulting nodes the recursively defined weights. All children of an internal node J have weight $w_J m_J$, so this is a Watanabe special datum. The converse part of Lemma 5.1 identifies the branching weight for i, j with d_{ij} . Lemma 5.2 therefore shows that this datum and $G_{\mathbf{p}}$ have exactly the same pair-supported subgroups. Both groups are generated by them, by (5.11) and Lemma 5.2; hence the two groups coincide. Formula (5.12) shows that the node monomials generate $\mathbb{C}[S(\mathbf{p})]$. They are nonredundant: if $w_J \mathbf{1}_J$ were a sum of other node vectors, every summand would have support contained

in J ; but every proper subnode $K \subsetneq J$ has $w_K > w_J$, so its nonzero coordinates already exceed those of $w_J \mathbf{1}_J$. Thus the node vectors are exactly the Hilbert basis. For each internal node there is the binomial identity

$$M_J^{m_J} = \prod_{K \prec J} M_K.$$

Watanabe's inductive proof [11, Proposition 1.7] gives exactly one such equation per internal node and proves (5.7). The quotient is the n -dimensional domain A , while the polynomial ring has $n + t(\mathbf{p})$ variables. Hence the $t(\mathbf{p}) = |\mathcal{D}| - n$ displayed binomials generate an ideal of height $t(\mathbf{p})$; since the polynomial ring is Cohen–Macaulay, they form a regular sequence. Since $X_i^{p_i} = Y_i^{c_i}$, adjoining F gives (5.8); F is regular because the toric quotient is a domain.

Every toric binomial lies in the square of the homogeneous maximal ideal. The last equation has a nonzero linear part exactly when some $c_i = 1$, and its linear part is then a sum of distinct leaf variables. Thus it removes exactly one cotangent-space generator. This proves (5.9).

The degree formulas in (5.10) follow directly from $N\vec{\tau} = \vec{c}$. Along the chain from a root of weight one to a leaf of weight q_i , the internal relation degrees telescope. Hence

$$\sum_{J \in \mathcal{D}^\circ} \deg(\text{relation at } J) + N - \sum_{J \in \mathcal{D}} \deg M_J = N - \sum_i \frac{N}{p_i} = \delta.$$

This is the complete-intersection Gorenstein parameter. A common divisor of all variable weights divides every relation degree, hence the displayed parameter δ , and it also divides N because $c_i \deg Y_i = N$. The coprimality $\gcd(\delta, N) = 1$ proves primitivity before eliminating a linear leaf. If $\epsilon = 1$, eliminate one such leaf and the last equation. Both have degree N , so their simultaneous elimination leaves the difference between the sum of relation degrees and the sum of variable degrees, and hence the Gorenstein parameter δ , unchanged. A common divisor of the remaining variable weights still divides all remaining relation degrees and hence δ ; it also divides $c_j \deg Y_j = N$ for any remaining leaf j . Since $n \geq 4$, such a leaf exists. Thus the minimal weights are primitive as well. $\square$

Corollary 5.4 (The hypersurface classification). *Under (5.1), the root ring is a hypersurface if and only if one of the following mutually exclusive alternatives holds.*

- (1) *The integers $p_1, \dots, p_n$ are pairwise coprime.*
- (2) *The gcd graph has exactly one nontrivial component C , and there are an integer $\lambda \geq 2$ and pairwise coprime positive integers c_i , $i \in C$, some $c_{i_0} = 1$, such that*

$$p_i = \lambda c_i \quad (i \in C).$$

Every exponent outside C is coprime to all the other exponents.

In the first case $R = T$. In the second, write the exponents outside C as $b_1, \dots, b_{n-|C|}$, and put

$$U = \prod_{i \in C} X_i, \quad Y_i = X_i^\lambda \quad (i \in C), \quad Z_j = X_j.$$

After eliminating Y_{i_0} , the root ring has the hypersurface presentation

$$(5.13) \quad \mathbb{C}[U, (Y_i)_{i \in C \setminus \{i_0\}}, Z_1, \dots, Z_{n-|C|}] / \left(U^\lambda + \prod_{i \neq i_0} Y_i \left(\sum_{i \neq i_0} Y_i^{c_i} + \sum_j Z_j^{b_j} \right) \right).$$

Its primitive weights and relation degree are

$$(5.14) \quad \deg U = \frac{N}{\lambda} \sum_{i \in C} \frac{1}{c_i}, \quad \deg Y_i = \frac{N}{c_i}, \quad \deg Z_j = \frac{N}{b_j}, \quad h = N \sum_{i \in C} \frac{1}{c_i}.$$

In particular, $h - \sum w_i = \delta$.

Proof. By (5.9), a hypersurface has either no internal forest node, or one internal node and $\epsilon = 1$. The first condition is pairwise coprimality. In the second case, the leaves below the unique node have a common weight $\lambda = q_i$; writing $p_i = \lambda c_i$, the normalized gcds of distinct children are one, so the c_i are pairwise coprime, while $\epsilon = 1$ says that some $c_i = 1$. All other leaves are singleton gcd components. Conversely, the data in (2) produce exactly this one-node forest. Eliminating the linear leaf Y_{i_0} from (5.8) gives (5.13); the degree formulas, primitivity, and Gorenstein parameter follow from (5.10). $\square$

Example 5.5 (A genuinely two-level root). The signature

$$(p_1, p_2, p_3, p_4) = (2, 4, 8, 11)$$

has $N = 88$, $\delta = 3$, and $(q_1, q_2, q_3, q_4) = (2, 4, 4, 1)$. Its nontrivial gcd component first splits with multiplier 2, and its child $\{2, 3\}$ splits once more with multiplier 2. With

$$U = X_1 X_2 X_3, \quad V = X_2^2 X_3^2, \quad Y_1 = X_1^2, \quad Y_2 = X_2^4, \quad Y_3 = X_3^4, \quad Z = X_4,$$

one obtains

$$R \cong \mathbb{C}[U, V, Y_1, Y_2, Y_3, Z]/(U^2 - Y_1 V, V^2 - Y_2 Y_3, Y_1 + Y_2 + Y_3^2 + Z^{11}).$$

Eliminating Y_1 gives the codimension-two presentation

$$R \cong \mathbb{C}[U, V, Y_2, Y_3, Z]/(U^2 + V(Y_2 + Y_3^2 + Z^{11}), V^2 - Y_2 Y_3).$$

The forest has two internal nodes, so this complete intersection is not a hypersurface and cannot be detected by the one-node classification.

6. THE ISOLATED LOCUS

We now identify and count the isolated hypersurfaces. No auxiliary family or bounded-index enumeration is used.

Proposition 6.1 (The isolatedness criterion). *Assume $n \geq 4$ and $R(\mathbf{p}, \vec{\tau})$ is a hypersurface. The following are equivalent:*

- (1) $\text{Spec } R(\mathbf{p}, \vec{\tau})$ has an isolated singularity;
- (2) $R(\mathbf{p}, \vec{\tau}) = T(\mathbf{p})$;
- (3) the integers p_i are pairwise coprime.

In this case $N = \prod_i p_i$ and

$$(6.1) \quad \sum_i \frac{1}{p_i} + \frac{\delta}{\prod_i p_i} = 1.$$

Proof. By Proposition 4.4, one has $u = 1$, and the rigid hypotheses (5.1) hold. Hence the semigroup description (5.3) applies. If the p_i are pairwise coprime, the Chinese remainder theorem gives $S(\mathbf{p}) = \mathbb{N}^n$ in (5.2); hence $R = T$, and the Brieskorn–Pham hypersurface has an isolated singularity.

Conversely, suppose $s = \gcd(p_i, p_j) > 1$ for some $i \neq j$. Since $n \geq 4$, the following locus is nonempty of dimension $n - 3$:

$$Z_{ij} = \{X_i = X_j = 0, X_\ell \neq 0 \ (\ell \neq i, j)\} \subset \text{Spec } T.$$

Indeed, the equation remaining in the $n - 2$ nonzero coordinates has solutions, and its gradient is nonzero there. We determine the stabilizer, rather than merely a subgroup of it. When $u = 1$, Proposition 3.1 gives

$$G_{\vec{\tau}} = \{(\alpha_l) : \alpha_l^{p_l} = 1, \prod_l \alpha_l = 1\}.$$

At a point $z \in Z_{ij}$, an element of the stabilizer has $\alpha_l = 1$ for $l \neq i, j$, and hence is uniquely of the form

$$(X_i, X_j) \mapsto (\zeta X_i, \zeta^{-1} X_j), \quad \zeta \in \mu_s.$$

Thus the stabilizer is exactly μ_s . Since a partial derivative in one of the remaining coordinates is nonzero at z , that coordinate may be eliminated formally, and

$$\widehat{\mathcal{O}}_{\text{Spec } T, z} \cong \mathbb{C}[[x_i, x_j, t_1, \dots, t_{n-3}]].$$

The completion of the quotient at the image of z is the invariant ring of this completion under its stabilizer, hence

$$\mathbb{C}[[t_1, \dots, t_{n-3}]] \widehat{\otimes} \mathbb{C}[[x_i, x_j]]^{\mu_s} \cong \frac{\mathbb{C}[[t_1, \dots, t_{n-3}, u, v, w]]}{(uv - w^s)}.$$

It is a smooth factor times an A_{s-1} singularity. The quotient therefore has a positive-dimensional singular locus and is not isolated. This expands the argument of [5, Proposition 3.2]. Equation (6.1) follows from (1.9). $\square$

For fixed n , the proposition converts the complete isolated locus into a pairwise-coprime lattice-point problem with a polynomial index function.

Theorem 6.2 (Full mean asymptotic on the isolated locus). *Fix $n \geq 4$, and let $I_n^{\text{iso}}(\delta)$ be the number of nondecreasing isolated hypersurface signatures of root index δ . Then*

$$(6.2) \quad \sum_{1 \leq \delta \leq X} I_n^{\text{iso}}(\delta) \sim C_n X (\log X)^{n-1},$$

where

$$(6.3) \quad C_n = \frac{\mathfrak{S}_n}{n!(n-1)!},$$

$$(6.4) \quad \mathfrak{S}_n = \prod_{\ell \text{ prime}} \left(1 - \frac{1}{\ell}\right)^{n-1} \left(1 + \frac{n-1}{\ell}\right) > 0.$$

The asymptotic has the following parity refinement. If n is even, then $I_n^{\text{iso}}(\delta) = 0$ for every even δ . If n is odd, then

$$(6.5) \quad \sum_{\substack{1 \leq \delta \leq X \\ \delta \text{ even}}} I_n^{\text{iso}}(\delta) \sim \frac{C_n}{n+1} X (\log X)^{n-1},$$

$$(6.6) \quad \sum_{\substack{1 \leq \delta \leq X \\ \delta \text{ odd}}} I_n^{\text{iso}}(\delta) \sim \frac{nC_n}{n+1} X (\log X)^{n-1}.$$

In particular $C_4 \approx 7.97806 \cdot 10^{-4}$. Consequently there is an unbounded sequence of indices δ_k such that

$$(6.7) \quad I_n^{\text{iso}}(\delta_k) \gg_n (\log \delta_k)^{n-1}.$$

If n is odd, such a sequence can be chosen separately in each parity class; if n is even, every isolated index is odd.

Proof. For an ordered n -tuple $\mathbf{p} = (p_1, \dots, p_n)$, put

$$(6.8) \quad P(\mathbf{p}) = \prod_{i=1}^n p_i, \quad \Delta(\mathbf{p}) = P(\mathbf{p}) \left(1 - \sum_{i=1}^n \frac{1}{p_i}\right).$$

If the p_i are pairwise coprime and $\Delta(\mathbf{p}) > 0$, then $\Delta(\mathbf{p})$ is a positive integer and

$$\Delta(\mathbf{p}) \equiv -\frac{P(\mathbf{p})}{p_i} \pmod{p_i}.$$

Thus $\gcd(\Delta(\mathbf{p}), P(\mathbf{p})) = 1$. The canonical-root criterion gives the maximal $\Delta(\mathbf{p})$ -th root, and Proposition 6.1 identifies its section ring with the isolated Brieskorn–Pham

hypersurface. Conversely every isolated signature arises in this way. Hence the left side of (6.2) is $1/n!$ times the number of ordered pairwise-coprime tuples in $\mathbb{Z}_{\geq 2}^n$ satisfying

$$(6.9) \quad 0 < \Delta(\mathbf{p}) \leq X.$$

The factor is exactly $1/n!$, because pairwise-coprime entries at least two are necessarily distinct.

We first record an elementary estimate which controls the cusps near an Egyptian-fraction boundary. For fixed positive integers A, B , define

$$F_{A,B,m}(x_1, \dots, x_m) = A \prod_{i=1}^m x_i - B \sum_{i=1}^m \prod_{j \neq i} x_j.$$

Then

$$(6.10) \quad \#\{\mathbf{x} \in \mathbb{Z}_{\geq 2}^m : 0 < F_{A,B,m}(\mathbf{x}) \leq X\} \ll_{A,B,m} X(1 + \log X)^{m-1}.$$

Indeed, this follows by induction on m . Choose a fixed integer $M > 2Bm/A$; the case $m = 1$ is immediate. If every $x_i \geq M$, then

$$F_{A,B,m}(\mathbf{x}) \geq \frac{A}{2} \prod_i x_i,$$

and the standard elementary divisor bound

$$\#\{\mathbf{x} \in \mathbb{Z}_{\geq 2}^m : \prod_i x_i \leq Y\} \ll_m Y(1 + \log Y)^{m-1}$$

applies. If $x_i = a < M$, then

$$(6.11) \quad F_{A,B,m}(\mathbf{x}) = F_{Aa-B, Ba, m-1}(x_1, \dots, \widehat{x_i}, \dots, x_m).$$

There are no positive values when $Aa - B \leq 0$; otherwise the induction hypothesis applies. There are only finitely many choices of i and a , which proves (6.10).

We next count ordered pairwise-coprime tuples by their ordinary product. This product-height input is classical: an ordered pairwise-coprime factorization of an integer m into n factors is obtained by assigning each prime-power divisor of m to one of the n coordinates, and is therefore counted by $n^{\omega(m)}$. Equivalently, allowing entries equal to one for the moment, the Dirichlet series is

$$(6.12) \quad D_n(s) = \sum_{\substack{p_1, \dots, p_n \geq 1 \\ (p_i, p_j) = 1 \ (i \neq j)}} \frac{1}{(p_1 \cdots p_n)^s}$$

$$(6.13) \quad = \prod_{\ell \text{ prime}} \left(1 + \frac{n\ell^{-s}}{1 - \ell^{-s}} \right) = \zeta(s)^n G_n(s),$$

$$(6.14) \quad G_n(s) = \prod_{\ell \text{ prime}} (1 - \ell^{-s})^{n-1} (1 + (n-1)\ell^{-s}).$$

The Euler product for G_n converges normally for $\Re(s) > 1/2$ in a neighborhood of $s = 1$, and $G_n(1) = \mathfrak{S}_n$. The Selberg–Delange theorem [8, Chapter II.5], applied to (6.13) therefore gives

$$(6.15) \quad \begin{aligned} & \#\{\mathbf{p} \in \mathbb{Z}_{\geq 1}^n : (p_i, p_j) = 1 \ (i \neq j), \ P(\mathbf{p}) \leq X\} \\ &= \frac{\mathfrak{S}_n}{(n-1)!} X(\log X)^{n-1} + O_n(X(\log X)^{n-2}). \end{aligned}$$

Requiring every p_i to exceed any fixed bound changes only the error term: after fixing one bounded coordinate, the remaining product count is $O_n(X(1 + \log X)^{n-2})$. Thus,

for every fixed M , the number $A_{n,M}(X)$ of ordered pairwise-coprime tuples with every $p_i \geq M$ and $P(\mathbf{p}) \leq X$ satisfies

$$(6.16) \quad A_{n,M}(X) \sim \frac{\mathfrak{S}_n}{(n-1)!} X(\log X)^{n-1}.$$

Let $H_n(X)$ denote the ordered count in (6.9), fix $M > n$, and first restrict to tuples with every $p_i \geq M$. On this region

$$\left(1 - \frac{n}{M}\right) P(\mathbf{p}) \leq \Delta(\mathbf{p}) < P(\mathbf{p}).$$

Consequently, if $H_{n,M}(X)$ denotes the restricted count, then

$$(6.17) \quad A_{n,M}(X) \leq H_{n,M}(X) \leq A_{n,M}\left(\frac{X}{1 - n/M}\right).$$

The complementary tuples have some $p_i = a < M$. After dropping all coprimality conditions, their remaining coordinates are counted by (6.10), because

$$\Delta(a, p_2, \dots, p_n) = F_{a-1, a, n-1}(p_2, \dots, p_n).$$

Thus, for fixed M ,

$$(6.18) \quad H_n(X) - H_{n,M}(X) = O_{n,M}\left(X(1 + \log X)^{n-2}\right).$$

Divide (6.17) by $X(\log X)^{n-1}$, use (6.16) and (6.18), and let first $X \rightarrow \infty$ and then $M \rightarrow \infty$. The lower and upper limits both equal $\mathfrak{S}_n/(n-1)!$. Dividing by $n!$ proves (6.2).

For the parity refinement, pairwise coprimality leaves two possibilities: all entries are odd, or exactly one entry is even. In the first case

$$\Delta(\mathbf{p}) \equiv 1 - n \pmod{2},$$

whereas in the second case $\Delta(\mathbf{p}) \equiv 1 \pmod{2}$. At the prime two, separate the product-height Dirichlet series as

$$\begin{aligned} D_n^{(0)}(s) &= \prod_{\ell \text{ odd}} \left(1 + \frac{n\ell^{-s}}{1 - \ell^{-s}}\right), \\ D_n^{(1)}(s) &= \frac{n2^{-s}}{1 - 2^{-s}} D_n^{(0)}(s), \end{aligned}$$

where the superscripts record respectively no even coordinate and exactly one even coordinate. Their local multipliers at $s = 1$ are 1 and n , whereas the full local multiplier is $n + 1$. Selberg–Delange therefore gives product-height leading constants

$$\frac{1}{n+1} \frac{\mathfrak{S}_n}{(n-1)!} \quad \text{and} \quad \frac{n}{n+1} \frac{\mathfrak{S}_n}{(n-1)!}.$$

Removing tuples with a coordinate below a fixed M changes only the $O_{n,M}(X(1 + \log X)^{n-2})$ term in either class. Apply (6.17) to each class separately; the omitted cusps are again bounded by (6.10), even after all parity conditions are dropped. Letting first $X \rightarrow \infty$ and then $M \rightarrow \infty$ transfers precisely these two constants to defect height. This proves (6.5) and (6.6), as well as the asserted vanishing when n is even.

Finally, averaging gives

$$\max_{\delta \leq X} I_n^{\text{iso}}(\delta) \geq (C_n + o(1))(\log X)^{n-1}.$$

The maximizing indices must be unbounded because every fixed-index multiplicity is finite by Theorem 1.2. Since $\delta \leq X$, an unbounded subsequence of maximizers gives (6.7). Applying the same argument to (6.5) and (6.6) gives the parity-refined assertion. $\square$

DATA AND CODE AVAILABILITY

The archive `higher_canonical_roots_v03_supplement.zip` contains three exact-arithmetic research programs, their recorded outputs, and one hardened orchestrator. The `gcd-forest` program reconstructs the laminar forest, its Hilbert basis, and its regular sequence in finite test boxes in four, five, and six variables. The isolated-mean program evaluates the four-variable isolated counting function and finite partial Euler products; the third program supplies independent CRT, forest, and literal-counting cross-checks. Every check uses an explicit exception and remains active under `python -O`, while the orchestrator also verifies the exact manifest. The file `HIGHER_CANONICAL_ROOTS_SUPPLEMENT_README_v03.md` records the reconstruction commands, software requirements, hashes, and MIT license. These computations are regression tests; no finite computation is used in the proofs.

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